

## **Project Title:**

AI-Enhanced Secure Enterprise Network with Real-Time Traffic Anomaly Detection

## **Project Overview:**

This project aims to design and implement a secure enterprise-level network integrated with an AI-based anomaly detection system. The network will follow a hierarchical architecture (access–distribution–core) with multiple VLAN-segmented subnets, incorporating both wired (Ethernet/fiber) and wireless components. In addition to standard networking functionality, the system will monitor live traffic and identify abnormal patterns such as port scanning, flooding, and unauthorized access attempts.

## **Objectives:**

The project will:

- Design a complete multi-layer enterprise network aligned with TCP/IP architecture
- Configure VLANs, inter-VLAN routing, trunking (802.1Q), and MAC learning behavior
- Implement IPv4/IPv6 addressing, subnetting, NAT/PAT, and ICMP-based diagnostics
- Deploy dynamic routing using OSPF and discuss MPLS concepts for scalability
- Establish secure communication using VPN (IPsec/WireGuard), TLS, and Zero-Trust-inspired access control
- Develop an AI-based module to detect anomalous traffic patterns
- Perform quantitative performance and security analysis under varying conditions

## **Methodology:**

The network will be simulated using Cisco Packet Tracer or GNS3. VLAN-based segmentation will be implemented across switches with trunk links, and both wired (CSMA/CD) and wireless (CSMA/CA) communication will be incorporated. Application-layer services including HTTP (web server), DNS resolution, and REST-based communication using sockets will be deployed.

Traffic will be generated using iPerf3 and captured via Wireshark. Extracted features such as packet rate, flow duration, and protocol distribution will be used to train a lightweight machine learning model (e.g., using Python and scikit-learn) to classify normal and malicious traffic.

Network monitoring will be implemented using SNMP, along with basic centralized logging to simulate SIEM-style observability.

## **Analysis Plan:**

The project will include quantitative evaluation of:

- Throughput, latency, packet loss, RTT, and jitter

- TCP vs UDP performance comparison under different network conditions
- Detection accuracy, precision, recall, and false positive rate of the AI model
- Performance overhead introduced by encryption, VPN, and anomaly detection

Comparative scenarios will include:

- Normal vs attack traffic
- Network performance with and without AI-based monitoring
- Secure vs non-secure configurations

Results will be presented using graphs and tables to support analytical conclusions.

### **Tools & Technologies:**

Cisco Packet Tracer / GNS3, Wireshark, iPerf3, Python (Scapy, scikit-learn), SNMP tools, and standard protocols (TCP/IP, OSPF, HTTP, DNS, TLS).

### **Expected Outcome:**

A fully functional enterprise network with integrated intelligent monitoring capable of detecting anomalous behavior in real time. The project will demonstrate practical implementation of networking, security, and AI concepts, along with a detailed evaluation of performance and security trade-offs.

### **Justification:**

Traditional enterprise networks rely on static rule-based security, which is insufficient against evolving threats. By integrating AI-driven anomaly detection with a structured and secure network design, this project provides a scalable and adaptive solution aligned with modern industry practices such as Zero-Trust Architecture and intelligent network monitoring.